

Week 5 - Lab

I. Prerequisite setup

1. Check where is your mysql located

```
SHOW VARIABLES LIKE '%dir%';
```

- basedir: Location of MySQL core files
- datadir: Location of database files

2. Check your type of storing engine

```
show variables like "%engine%";
```

3. Check if log bin are enable

```
show variables like "%log_bin%";
```

- if not you need to edit the mysql config file to enable.

```
[mysqld]  
...  
server-id=1  
log_bin=mysql-bin  
binlog_format=ROW  
expire_logs_days=7
```

4. Create one database called **big_data_db**

5. Install python

6. Download the script within this [link](#) and test running it to check if data being insert in your db or not.

II. Cold Backup/Recovery Exercise

1. Logical backup

You are going to perform logical backup on **airline** database.

```
## backup all database within DBMS
mysqldump -u root -p --all-databases > full_backup.sql
## backup specific database (Exclude create database statement)
mysqldump -u username -p [db] > database_backup.sql
## backup specific databases (Include create database statement)
mysqldump -u username -p --databases [db1 db2] > database_backup.sql
## Backup Selected Tables from a Database
mysqldump -u username -p [db] [tb1 tb2] > tables_backup.sql
## Backup Specific Rows
mysqldump -u username -p [db] [tb] --where="condition" > row-backup.sql
```

2. Perform drop database

3. Perform restoration

```
## if no create database statement
mysql -u username -p [db] < database_backup.sql
## if create database statement exist
mysql -u username -p < database_backup.sql
```

III. Hot Backup/Recovery Exercise

You are going to perform a logical backup on this **big_data_db** database while script are running.

While performing backup on running environment where data flow in every second, without a proper backup technique, you can loss some data.

We also need to do Transactional Log backup to simulate differential/incremental backup.

1. Perform Logical full backup

```
mysqldump -u root -p \  
--single-transaction \  
--flush-logs \  
--master-data=2 \  
big_data_db > full_big_data_db.sql
```

--single-transaction : *take a **consistent snapshot of InnoDB tables using a transaction**, without locking tables.*

`--flush-logs` : rotates binlog file by starts a new binlog file for incremental tracking.

```
--master-data=2 : injects metadata into dump file
```

```
## Show which binlog are being write.
SHOW MASTER STATUS; # for mairadb
SHOW BINARY LOG STATUS; # for mysql
## list all binlog.
SHOW BINARY LOGS;
```

2. Perform Transactional Incremental backup

first rotate log to write on new file, then perform Transactional Incremental backup.

```
## Rotate log to write on new log file
flush logs;
```

```
# Perform transactional incremental backup
mysqlbinlog --database=big data db \
```

```
mysql-bin.000002 > incr_002.sql
```

3. Perform Point-in-time-recovery (PITR)

The backup flow are: `full_big_data_db` $\Rightarrow$ `incr_002` $\Rightarrow$ binlog

You are going to perform a PITR to the earliest time before script finished 20 sec (scenario: software error in the last 20 sec of running).

```
## drop and create new "big_data_db"
mysql -u root -p big_data_db < full_big_data_db.sql
mysql -u root -p big_data_db < incr_002.sql
# todo: do backup to the time before error (--stop-datetime),
# run last restoration.
```

IV. Review Question

1. What is the difference between a physical backup and a logical backup in MySQL? Compare their advantages, disadvantages.
2. Why is the `--single-transaction` option important when performing a hot backup on an active database? Explain how it works with InnoDB tables and what problems might occur if it is not used.
3. What is the purpose of MySQL Binary Logs (Binlogs)? Describe what information is stored in binlogs and how they support incremental backup, replication, and point-in-time recovery.
4. How does Point-in-Time Recovery (PITR) work in MySQL? Explain the recovery process using a full backup and binary logs. Include the role of `-stop-position` and `-stop-datetime`.
5. What are the limitations of using `mysqldump` for backing up large production databases?

V. Submission

- Do research and answer all 5 questions.
- Compile the answer into pdf and submit on LMS.